Hongyi **Du**

Research Assistant · Undergraduate Researcher

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Education

University of Illinois at Urbana-Champaign *Illinois, United States*

B.S. in Computer Science; GPA:3.83/4.00 *Sep. 2022 - May. 2026*

- Achieved top grades (A+, A) in rigorous coursework: Advanced Natural Language Processing, Computational Photography, System programming, Algorithms, Numerical Methods, Computer Architecture, Data Structures, Probability and Statistics, and Linear Algebra. **Especially, Earn prefect A+(100/100) in Graduate course Advanced Natural Language Processing.**

Research Interests

LLM agents and tool-augmented reasoning; multi-agent communication and protocol selection; self-improving / self-evolving agent systems; language-agent workflows for scientific modeling and decision support;

Publications

Which LLM Multi-Agent Protocol to Choose?

Hongyi Du, Jiaqi Su, Jisen Li, Lijie Ding, Yingxuan Yang, Peixuan Han, Xiangru Tang, Kunlun Zhu, Jiaxuan You *Sept. 19, 2025 (revised: Sept. 28, 2025)*

- **Status:** under review at ICLR 2026.
- **arXiv:** arxiv.org/abs/2510.17149

ModelingAgent: Bridging LLMs and Mathematical Modeling for Real-World Challenges

Cheng Qian, **Hongyi Du**, Hongru Wang, Xiusi Chen, Yuji Zhang, Avirup Sil, Chengxiang Zhai, Kathleen McKeown, Heng Ji *May 2025*

- **arXiv:** arxiv.org/abs/2505.15068
- **EMNLP 2025 Findings**

MultiAgentBench: Evaluating the Collaboration and Competition of LLM Agents

Kunlun Zhu, **Hongyi Du**, Zhaochen Hong, Xiaocheng Yang, Shuyi Guo, Zhe Wang, Zhenhailong Wang, Cheng Qian, Xiangru Tang, Heng Ji, Jiaxuan You *Mar. 2025*

- **arXiv**: arxiv.org/abs/2503.01935
- **ACL 2025 Main Conference**

EscapeBench: Towards Advancing Creative Intelligence of Language Model Agents

Cheng Qian, Peixuan Han, Qinyu Luo, Bingxiang He, Xiusi Chen, Yuji Zhang, **Hongyi Du**, Jiarui Yao, Xiaocheng Yang, Denghui Zhang, Yunzhu Li, Heng Ji *Dec. 2024*

- **arXiv**: arxiv.org/abs/2412.13549
- **ACL 2025 Main Conference**

Research Experience

Bio-Inspired Multi-Agent Evolution Arena: Civilization Simulation and Swarm Intelligence *Champaign, United States*

Undergraduate Research Assistant at Blender Lab & U Lab — supervised by Prof. Gene Robinson, Prof. Heng Ji, and Prof. Jiaxuan You *Jun. 2025 – Present*

- **Content:** Build a bio-inspired multi-agent arena where civilizations evolve under limited resources, seasonal changes, and stochastic disasters.
- **Method:** Formalize self-evolving civilizations via mutation, inheritance, and adaptation loops over agent roles, communication protocols, memory, and tools, with group-level performance as the main selection signal.
- **My role:** Lead the overall scenario design and environment implementation (e.g., grid world, resource dynamics, disaster events), and coordinate interfaces with observation/analysis modules.

Implicit Task Affordance Prediction in 3D Space (Language-Conditioned PointFormer) *Remote (Evanston, United States)*

Remote Researcher at Northwestern University — supervised by Prof. Manling Li; mentored by Qineng Wang *Oct. 2025 – Present*

- **Content:** Study navigation-focused 3D affordance prediction that links videos, point clouds, and camera trajectories to implicit natural language task instructions.
- **Method:** Inject simple linguistic priors into PointFormer-style backbones and build a data pipeline where VLMs annotate videos with per-frame subtasks and affordances, then project them into 3D space.
- **My role:** Design the language-conditioning component and early data curation pipeline; lead the transition from VLM-only prototypes to a navigation model conditioned on language and affordance signals.

ProtocolBench & ProtocolRouter: Benchmarking and Dynamic Routing for LLM Multi-Agent Communication

Champaign, United States

Undergraduate Research Assistant at U Lab (UIUC) — supervised by Prof. Jiaxuan You *May 2025 – Oct. 2025*

- **Content:** Develop ProtocolBench, a protocol-agnostic benchmark for LLM-based multi-agent communication (A2A, ACP, ANP, Agora) and ProtocolRouter, a scenario-aware selector that chooses one protocol per module.
- **Method:** Build a three-layer evaluation stack (protocol adapters, scenario harness, metrics/logger) and design representative scenarios (GAIA QA, safety tech, streaming queue, fail-storm recovery) to jointly evaluate task utility, latency, resilience, and security.
- **My role:** Act as team lead and first author, responsible for benchmark design, scenario specification, analysis of protocol behaviors, and the end-to-end implementation and training of ProtocolRouter.
- **Outcome:** Show that learned routing achieves 63.3% scenario / 81.7% module selection accuracy and matches or surpasses the best single-protocol baselines (e.g., lower mean latency than ACP in streaming, faster recovery than A2A in fail-storm). Code: github.com/ulab-uiuc/AgentProtocols. **Manuscript under double-blind review at ICLR 2026**.

MultiAgentBench: Evaluating the Collaboration and Competition of LLM Agents

Champaign, United States

Undergraduate Research Assistant at Blender Lab — supervised by Prof. Heng Ji and Prof. Jiaxuan You *Aug. 2024 – Feb. 2025*

- **Content:** Develop MultiAgentBench, a benchmark for LLM-based multi-agent systems across coding, games (e.g., Minecraft), and research-style tasks, focusing on planning, memory, and collaborative abilities.
- **Method:** Design the Werewolf Arena framework to simulate decentralized social interactions and evaluate theory-of-mind reasoning, leadership, cooperation, and emergent distrust among LLM agents.
- **My role:** Co-first author; independently build the Werewolf environment, logging and analysis pipeline, and perform detailed behavioral analysis of multi-agent interactions. Discovered abnormal conservation strategy of LLM in Werewolf environment.
- **Outcome:** Reveal that stronger foundation models tend to develop mutual distrust and internal friction in social deduction games. Accepted by **ACL 2025 Main Conference**. Code: github.com/MultiagentBench/MARBLE | arXiv: 2503.01935.

ModelingBench & ModelingAgent: Real-World Mathematical Modeling Benchmark

Champaign, United States

Undergraduate Research Assistant at Blender Lab — supervised by Prof. Heng Ji *Feb. 2025 – Aug. 2025*

- **Content:** Construct a benchmark and agent framework for real-world mathematical modeling tasks (e.g., traffic, renewables, trading, ecosystem planning).
- **Method:** Develop DataAgent and SimulationAgent to automate data acquisition and quantitative simulation, and build reusable baseline workflows/tools for various base models across 60+ tasks.
- **My role:** Second author; lead core agent modules and new task curation (expanding coverage by ∼40%), and co-design multi-dimensional evaluation metrics and logging.
- **Outcome:** Provide a unified evaluation suite for LLM-based modeling agents on realistic decision problems. Submitted to **EMNLP 2025 Findings**. Code: github.com/qiancheng0/ModelingAgent | arXiv: 2505.15068.

ProfileBench: Personalized Dialogue Benchmark

Champaign, United States

Undergraduate Research Assistant at Blender Lab — supervised by Prof. Heng Ji *Aug. 2024 – Dec. 2024*

- **Content:** Build a benchmark to test whether LLMs can conduct engaging, personalized dialogue grounded in user profiles over long-horizon conversations.
- **Method:** Use GPT-4o with tools to generate realistic profile–dialogue pairs with evolving user states, and design evaluation protocols for profile grounding, memory, and engagement.
- **My role:** Implement the benchmark pipeline, construct the profile/dialogue dataset, and design metrics for profile adaptation and long-term recall.
- **Outcome:** Provide an internal benchmark for profiling and long-context personalization, used to guide subsequent Blender Lab projects on personalized LLM agents.

Project Experiences

CS598JH: Text Mining with LLMs — Novelty Detection for LLM-Generated Content

Champaign, United States

Researcher — supervised by Prof. Jiawei Han; mentored by Heng Wang *Aug. 2025 – Present*

- **Content:** Study how to detect novelty in LLM-generated answers under RAG settings, distinguishing truly new content from restatements or simple recombinations of retrieved knowledge.
- **Method:** Propose several complementary novelty dimensions (e.g., restatement, recombination, genuinely new ideas) and design annotation schemes that remain stable under heavy paraphrasing.
- **My role:** Lead the design and implementation of an originality-check pipeline that flags outputs whose wording differs significantly from retrieved evidence but whose semantics are nearly identical, and down-weights such pseudo-novel content.
- **Outcome:** Apply the framework to long-form ML research proposals to jointly analyze originality and quality under different RAG configurations. Ongoing course-to-paper project for CS598JH, targeting submission to **ACL**.

CS497: Individual Research — Knowledge-Graph-Enhanced Retrieval-Augmented Generation

Champaign, United States

Researcher — supervised by Prof. Jiawei Han *Jul. 2024 – Feb. 2025*

- **Content:** Explore replacing unstructured text corpora with knowledge graphs (KGs) as retrieval sources in RAG systems for complex QA tasks.
- **Method:** Test different KG construction strategies and propose a two-stage paradigm: first retrieve relevant article summaries from a large database, then construct a task-specific KG for each QA instance to support reasoning.
- **My role:** Design and implement the KG-based retrieval pipeline, run comparative experiments against text-only RAG baselines, and analyze when structured retrieval yields gains on knowledge-intensive questions.